

ASESII 24A02 Project-Based Quiz 1

Ridge, Lasso, and Regularization for Neural Features

Group 01:

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July 14, 2026

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Authorship and Contribution Statement

We, the undersigned members of **Group 01**, confirm that every member is able to explain any part of this analysis when asked. All members participated in the design, implementation, and review of this report. Any AI-assisted work is documented in the AI Usage Log (Appendix [A](#)).

Table 1: Individual contributions and verification responsibilities.

Student	ID	Main Contributions	Verification Performed
Nguyễn Đình Thiên Lộc	24125093	<i>TODO</i>	<i>TODO</i>
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Abstract

This report investigates regularised regression methods for predicting body-fat percentage (brozek) from anthropometric measurements in the `fat.csv` dataset ($n = 252$, $p = 14$ candidate predictors). After excluding leakage-prone variables, we compare Ordinary Least Squares (OLS), Ridge, Lasso, and Elastic Net regression using 5-fold cross-validation. We further explore fixed random ReLU feature expansions to bridge classical penalisation with neural-network concepts.

1 Prediction Design and Leakage Control

1.1 Target and Predictors

The `fat.csv` dataset contains $n = 252$ observations and 18 variables describing body composition and anthropometric measurements of adult males. The response variable is `brozek`, the body-fat percentage estimated via Brozek’s equation.

After inspection, the following variables are **excluded** from the predictor set to prevent data leakage:

- `siri` — an alternative body-fat estimate; including it would constitute a near-perfect proxy for the response.
- `density` — body density is the direct input to both the Brozek and Siri formulas; using it makes prediction trivially deterministic.
- `free` — fat-free weight is algebraically derived from body fat and total weight, creating a circular dependency.

This leaves $p = 14$ candidate predictors (age, weight, height, adipos, neck, chest, abdom, hip, thigh, knee, ankle, biceps, forearm, wrist).

Summary statistics. Table 2 reports the descriptive statistics for all retained predictors computed on the *training set only*.

Table 2: Descriptive statistics of retained predictors (training set).

tab_summary_stats.tex — run the R pipeline to generate.

1.2 Split and Train-Only Preprocessing

The data are split into a training set ($n_{\text{train}} \approx 201$, $\sim 80\%$) and a held-out test set ($n_{\text{test}} \approx 51$, $\sim 20\%$) using a fixed random seed for reproducibility.

Standardisation. All predictors are centred and scaled using the *training-set* mean and standard deviation. The same transformation is then applied to the test set to prevent information leakage from the test data into the modelling pipeline. Formally, for each predictor j :

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j^{\text{train}}}{s_j^{\text{train}}}, \quad i = 1, \dots, n.$$

Cross-validation. We use $K = 5$ -fold cross-validation (CV) throughout, with folds assigned once and reused across all models to ensure fair comparison.

1.3 Training-Data Exploration

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Run the R pipeline first.

Figure 1: Pearson correlation heatmap of training-set predictors and response.

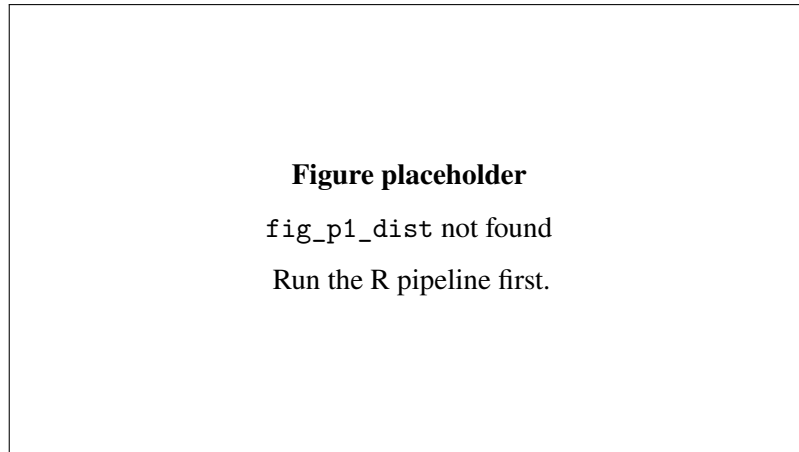


Figure 2: Distribution of the response variable brozek in the training set.

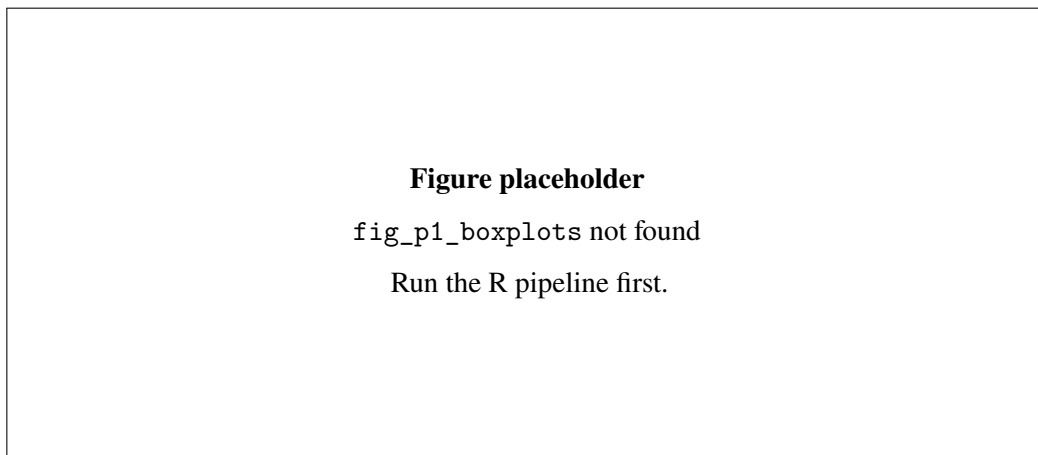


Figure 3: Side-by-side boxplots of standardised predictors (training set).

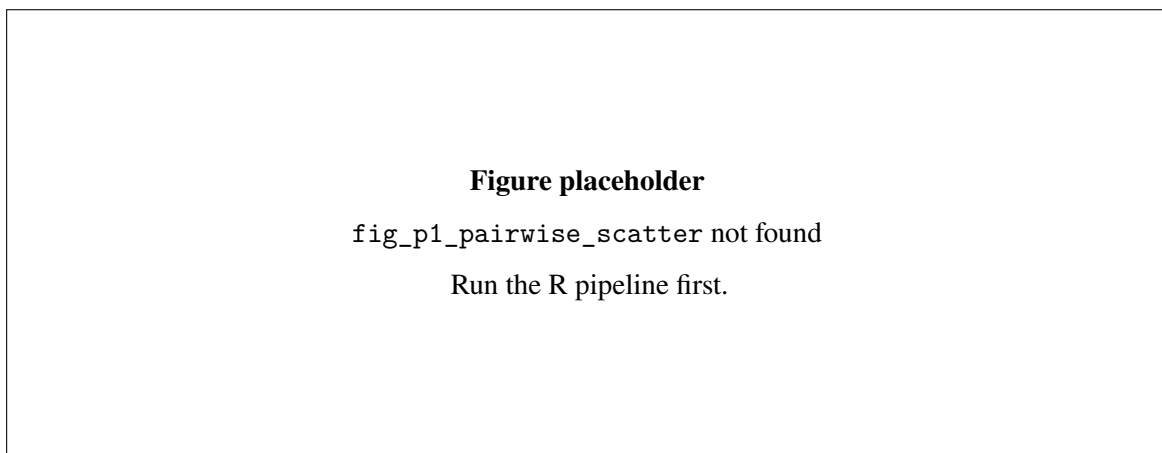


Figure 4: Pairwise scatter plots of selected predictors against brozek.

2 OLS, Ridge, and Lasso Regression

2.1 OLS Baseline (P2 — Tuấn)

Ordinary Least Squares provides the unpenalised baseline:

$$\hat{\beta}^{\text{OLS}} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Table 3: OLS coefficient estimates on the training set.

`tab_ols_coef.tex` — run the R pipeline to generate.

Figure placeholder

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Run the R pipeline first.

Figure 5: OLS diagnostic plots: residuals vs. fitted values and Q-Q plot.

2.2 Ridge Regression (P2 — Tuấn)

Ridge regression adds an ℓ_2 penalty to the OLS objective:

$$\hat{\beta}^{\text{Ridge}} = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}.$$

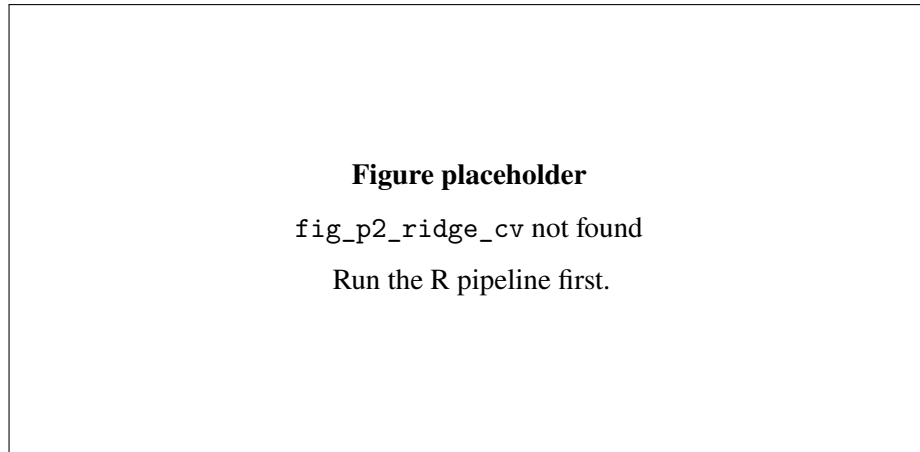


Figure 6: Ridge regression: cross-validated MSE as a function of $\log \lambda$.

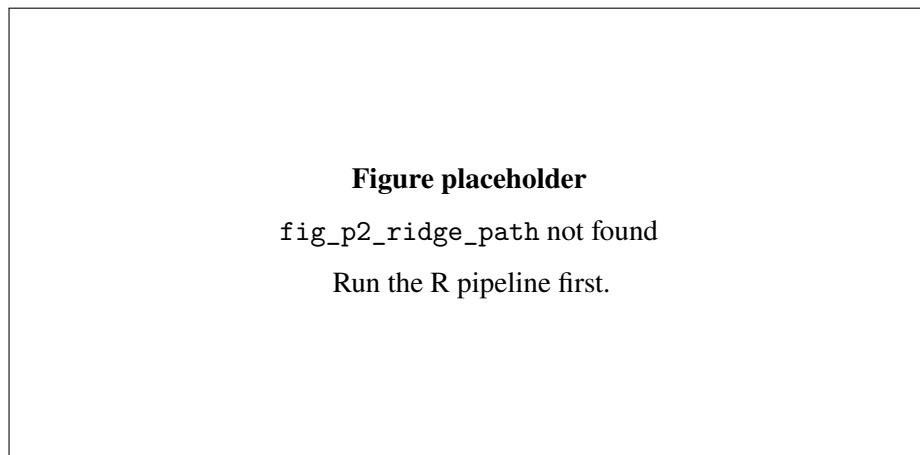


Figure 7: Ridge coefficient path: coefficients vs. $\log \lambda$.

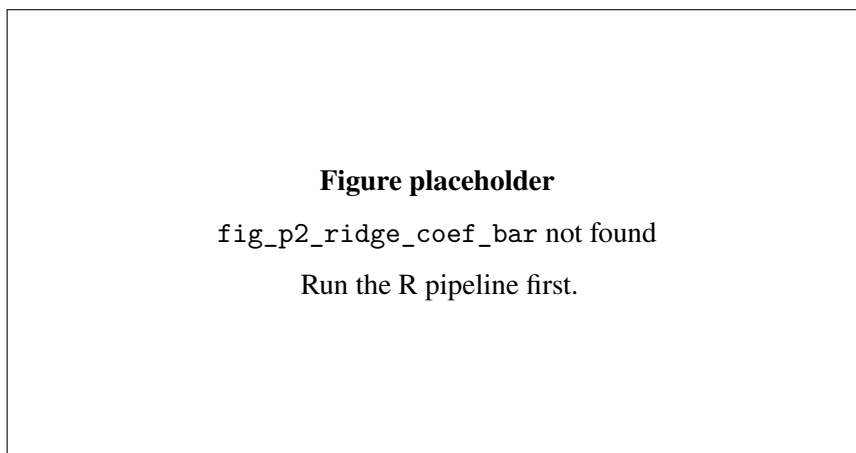


Figure 8: Ridge coefficient estimates at $\lambda_{\min}$ vs. λ_{1se} .

Table 4: Condition number of $\mathbf{X}^\top \mathbf{X}$ vs. $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ for selected λ values.

`tab_condition_numbers.tex` — run the R pipeline to generate.

Tuning results.

2.3 Lasso Regression (P3 — Thuận)

The Lasso replaces the ℓ_2 penalty with an ℓ_1 penalty, enabling automatic feature selection:

$$\hat{\boldsymbol{\beta}}^{\text{Lasso}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}.$$

Figure placeholder

`fig_p2_lasso_cv` not found

Run the R pipeline first.

Figure 9: Lasso regression: cross-validated MSE as a function of $\log \lambda$.

Figure placeholder

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Run the R pipeline first.

Figure 10: Lasso coefficient path: coefficients vs. $\log \lambda$.

Table 5: Lasso-selected features and their coefficient estimates.

`tab_lasso_selected.tex` — run the R pipeline to generate.

2.4 Training-Based Comparison and Core Model (P3 — Thuận)

Table 6: Training CV performance comparison across OLS, Ridge, and Lasso.

`tab_model_comparison.tex` — run the R pipeline to generate.

Figure placeholder
`fig_p2_comparison` not found
Run the R pipeline first.

Figure 11: Cross-validated RMSE comparison: OLS vs. Ridge vs. Lasso.

Core Model Declaration (pre-holdout):

TODO: Declare and justify core model choice here.

2.5 Perturbation or Stability Check (P3 — Thuận)

Prediction before check. *TODO: State prediction.*

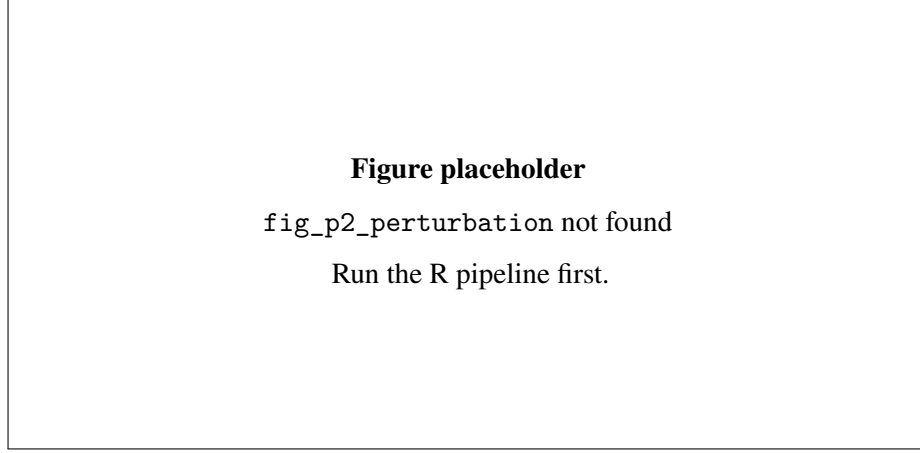


Figure 12: Coefficient stability under predictor perturbation or bootstrap resampling.

3 Mathematical Mechanisms

3.1 Ridge Regression and Conditioning

Definition 3.1 (Ridge Objective). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the design matrix, $\mathbf{y} \in \mathbb{R}^n$ the response vector, and $\boldsymbol{\beta} \in \mathbb{R}^p$ the coefficient vector. The Ridge regression objective is:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \{ \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \}, \quad \lambda > 0. \quad (1)$$

Derivation. Expanding the objective function:

$$\begin{aligned} L(\boldsymbol{\beta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta} \\ &= \mathbf{y}^\top \mathbf{y} - 2 \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta}. \end{aligned} \quad (2)$$

Taking the gradient with respect to $\boldsymbol{\beta}$ and setting it to zero:

$$\begin{aligned} \frac{\partial L}{\partial \boldsymbol{\beta}} &= -2 \mathbf{X}^\top \mathbf{y} + 2 \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} + 2\lambda \boldsymbol{\beta} = \mathbf{0} \\ \implies (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \boldsymbol{\beta} &= \mathbf{X}^\top \mathbf{y}. \end{aligned} \quad (3)$$

Since $\mathbf{X}^\top \mathbf{X}$ is positive semi-definite and $\lambda > 0$, the matrix $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p$ is **strictly positive definite**, hence always invertible. The unique closed-form solution is:

$$\boxed{\hat{\boldsymbol{\beta}}^{\text{Ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{y}.} \quad (4)$$

Remark 3.2 (Conditioning). If $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p > 0$ are the singular values of $\mathbf{X}$, then the condition number of $\mathbf{X}^\top \mathbf{X}$ is $\kappa(\mathbf{X}^\top \mathbf{X}) = (\sigma_1/\sigma_p)^2$. Adding the ridge penalty changes the eigenvalues to $\sigma_j^2 + \lambda$, so:

$$\kappa(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) = \frac{\sigma_1^2 + \lambda}{\sigma_p^2 + \lambda} \leq \frac{\sigma_1^2}{\sigma_p^2} = \kappa(\mathbf{X}^\top \mathbf{X}).$$

As λ increases, the condition number decreases monotonically toward 1, stabilising the inversion.

3.2 Lasso Optimality and Soft Thresholding

Definition 3.3 (Lasso Objective).

$$\hat{\beta}^{\text{Lasso}} = \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1 \right\}. \quad (5)$$

Subgradient optimality conditions. Because $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ is not differentiable at $\beta_j = 0$, we use subgradient calculus. The subgradient optimality condition for coordinate j is:

$$-\frac{1}{n} \mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\beta}) + \lambda \partial |\hat{\beta}_j| \ni 0, \quad (6)$$

where $\partial |\hat{\beta}_j|$ is the subdifferential of the absolute value:

$$\partial |\hat{\beta}_j| = \begin{cases} \{+1\} & \text{if } \hat{\beta}_j > 0, \\ \{-1\} & \text{if } \hat{\beta}_j < 0, \\ [-1, +1] & \text{if } \hat{\beta}_j = 0. \end{cases}$$

Special case: orthonormal design. When $\frac{1}{n} \mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$, each coordinate decouples and the Lasso admits the closed-form **soft-thresholding** solution:

$$\boxed{\hat{\beta}_j^{\text{Lasso}} = \text{sign}(\hat{\beta}_j^{\text{OLS}}) \cdot \max(|\hat{\beta}_j^{\text{OLS}}| - \lambda, 0)}. \quad (7)$$

This means:

- If $|\hat{\beta}_j^{\text{OLS}}| > \lambda$: the coefficient is *shrunk* toward zero by exactly λ .
- If $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda$: the coefficient is set *exactly to zero* — the variable is dropped.

3.3 Connecting Algebra to Evidence

TODO: Provide a concrete numerical example linking the theory from Sections 3.1–3.2 to the empirical results from Section 2.

4 Elastic Net and Neural-Feature Bridge

4.1 Research Question and Source Map

TODO: State research question here.

Table 7: Source map linking claims to evidence.

Claim	Source	Establishes	Does Not Establish
<i>TODO: Claim 1</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>
<i>TODO: Claim 2</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>
<i>TODO: Claim 3</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>

4.2 Elastic Net on Original Predictors

The Elastic Net [Zou and Hastie, 2005] combines ℓ_1 and ℓ_2 penalties:

$$\hat{\beta}^{\text{ENet}} = \arg \min_{\beta} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right] \right\}, \quad \alpha \in [0, 1].$$

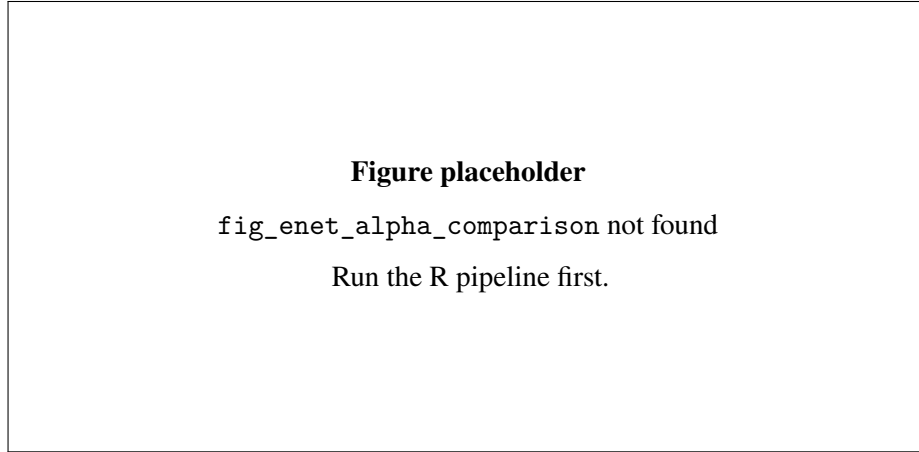


Figure 13: Elastic Net CV error across different α values.

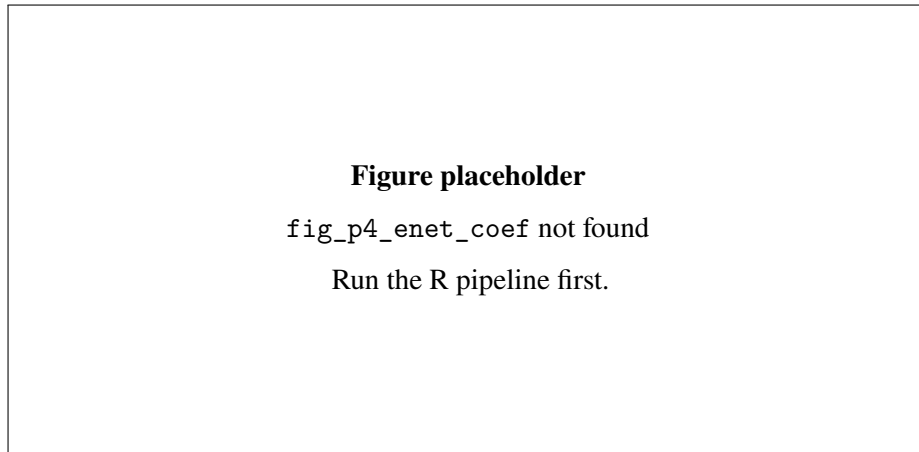


Figure 14: Elastic Net coefficient estimates at optimal (α, λ) .

4.3 Fixed ReLU Features

To bridge regularised regression with neural-network concepts, we generate $M = 200$ random features via:

$$h_m(x) = \max(w_m^\top x + b_m, 0), \quad m = 1, \dots, M,$$

where $w_m \sim \mathcal{N}(\mathbf{0}, I_p)$ and $b_m \sim \mathcal{N}(0, 1)$ are **fixed** (not learned). Only the output-layer coefficients $\beta \in \mathbb{R}^M$ are learned via Ridge/Lasso/Elastic Net on the transformed design matrix $\mathbf{H} \in \mathbb{R}^{n \times M}$.

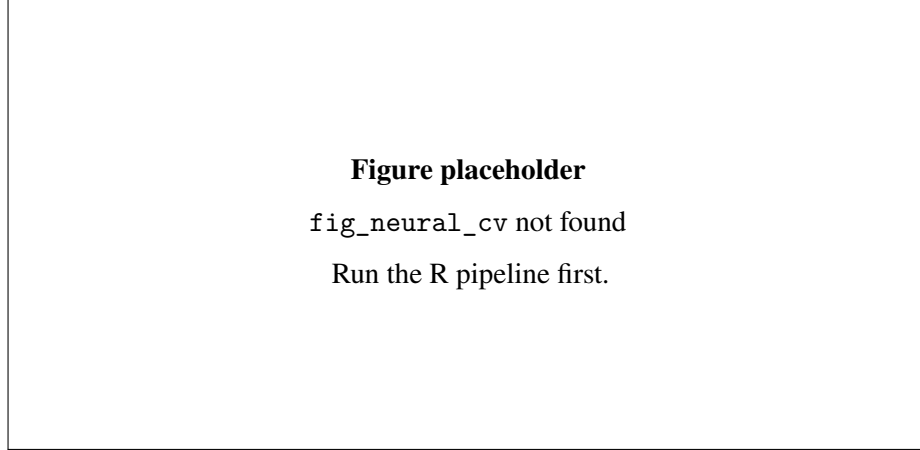


Figure 15: CV error of Ridge/Lasso/Elastic Net on fixed ReLU features ($M = 200$).

4.4 Locked Declaration and Final Holdout

Locked Model Declaration

Feature set: *TODO: original 14 / ReLU 200 / other*

Method: *TODO: Ridge / Lasso / Elastic Net*

Tuning: *TODO: $\alpha = \dots$, $\lambda = \dots$*

Justification: *TODO: brief reason*

This declaration was made before examining holdout results.

Table 8: Final holdout performance: RMSE and R^2 on the test set.

tab_holdout_comparison.tex — run the R pipeline to generate.

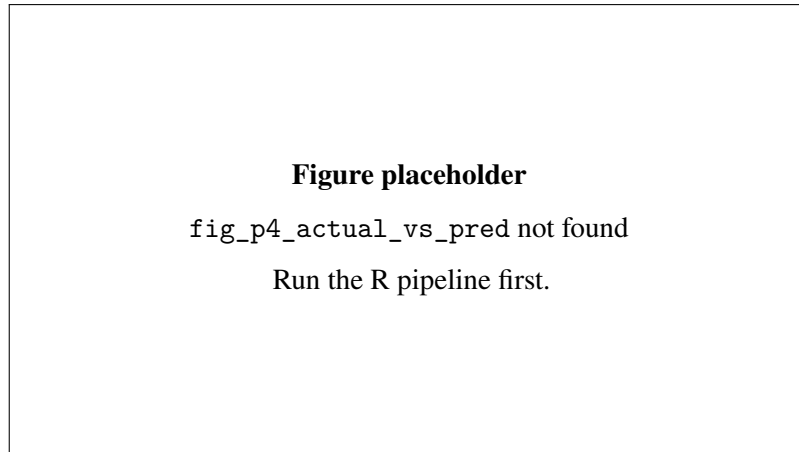


Figure 16: Actual vs. predicted brozek on the holdout test set.

4.5 Deep-Learning Connection and Limitations

TODO: Write the deep-learning connection discussion here.

Limitations. *TODO: Discuss ≥ 2 limitations.*

5 Report Quality and Reproducibility

5.1 Reproduction Record

The following step-by-step instructions reproduce all results, figures, and tables in this report from scratch.

Prerequisites.

- R $\geq 4.3.0$ with packages: glmnet, ggplot2, dplyr, corrplot, caret, knitr, xtable.
- L^AT_EX distribution (e.g. TeX Live 2024 or MiKTeX) with latexmk.
- The dataset fat.csv placed in project_quiz1_latex/data/.

Steps.

```
1 # 1. Clone or download the project
2 cd D:/Stats2/.
3
4 # 2. Run the R analysis pipeline
5 make all
6 # This generates all figures in output/figures/
7 # and all tables in output/tables/
8
9 # 3. Compile the LaTeX report
10 cd report
11 latexmk -pdf main.tex
12
13 # 4. Clean auxiliary files (optional)
14 latexmk -c
```

Listing 1: Reproduction commands.

5.2 Recommendation and Limitations

Recommendation. *TODO: State your final model recommendation with supporting evidence from the cross-validation and holdout results.*

Prediction vs. Interpretation. *TODO: Discuss the trade-off. Ridge retains all predictors (better for interpretation of relative importance), while Lasso provides a sparse model (easier to deploy and communicate). Elastic Net offers a middle ground.*

Limitations.

1. *TODO: Limitation 1 (e.g., sample size and generalisability).*
2. *TODO: Limitation 2 (e.g., linearity assumption).*
3. *TODO: Limitation 3 (e.g., predictor measurement error).*

5.3 AI Verification Record

All AI-assisted outputs were independently verified by team members. Table 9 summarises the verification process.

Table 9: AI verification record.

AI-Assisted Item	How Checked	Evidence	Modification
<i>TODO: Item 1</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>
<i>TODO: Item 2</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>
<i>TODO: Item 3</i>	<i>TODO</i>	<i>TODO</i>	<i>TODO</i>

References

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005. doi: 10.1111/j.1467-9868.2005.00503.x.

A AI Usage Log

This appendix documents all instances where AI tools (e.g., ChatGPT, GitHub Copilot, Gemini) were used during the preparation of this report. The full log is maintained in `Group01_AI_Log.csv`.

#	Date	AI Tool & Prompt Summary	Output Used For	Human Verification
1	TODO	TODO: Describe the prompt	TODO: What was the output used for?	TODO: How was it verified?
2	TODO	TODO	TODO	TODO
3	TODO	TODO	TODO	TODO

Table 10: AI usage log for Group 01.